

Weekly Notice 4

Exercises, Week 4:

1. Exercise 35.
2. Exercise 39.
3. Exercise 40.
4. Exercise 33.
5. Exercise 7.
6. Exercise 60.

Misprint: In (4.33) the Q 's are written in reversed order. Change to

$$\mathbb{P}_x(X_1 \in B_1, \dots, X_n \in B_n) = \int_{B_1} \cdots \int_{B_n} \mathbb{1} Q(y_{n-1}, dy_n) \cdots Q(y_1, dy_2) Q(x, dy_1).$$

In addition, it is worth emphasizing the following result, which is a consequence of (4.35):

For $n \in \mathbb{N}$ and $i, i_1, \dots, i_n \in S$,

$$\begin{aligned} \mathbb{P}_i(X_1 = i_1, \dots, X_n = i_n) &= \mathbb{P}_i(X_0 = i, X_1 = i_1, \dots, X_n = i_n) \\ &= \delta_i(\{i\}) p_{ii_1} \cdots p_{i_{n-1}i_n} \\ &= p_{ii_1} \cdots p_{i_{n-1}i_n}. \end{aligned}$$

An alternative proof for (4.29) in Proposition 4.25: Note that

$$\mathbb{E}_\mu[Z] = \mathbb{E}_\mu[\mathbb{E}_\mu[Z \mid \mathcal{F}_0^X]] = \mathbb{E}_\mu[\mathbb{E}_\mu[Z \circ \theta^0 \mid \mathcal{F}_0^X]],$$

where in the last equality, we use that θ^0 is the identity map. Now, using Theorem 4.27(1) followed by the Transformation Theorem, it follows that the above is also equal to

$$\mathbb{E}_\mu[\mathbb{E}_{X_0}[Z]] = \int_S \mathbb{E}_x[Z] (\mathbb{P}_\mu)_{X_0}(dx).$$

Here, $(\mathbb{P}_\mu)_{X_0}$ denotes the distribution of X_0 under $\mathbb{P}_\mu$, which is μ . Thus, we obtain

$$E_\mu[Z] = \int_S \mathbb{E}_x[Z] (\mathbb{P}_\mu)_{X_0}(dx) = \int_S \mathbb{E}_x[Z] \mu(dx)$$

as desired.